

READING LIST: SPAN 589

Title: Data Science for Linguists

01:940:589:01 - Spring 2021

Meetings: via Zoom, 08:10–11:10

Professor: Joseph V. Casillas, PhD

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Office hours: by appointment

Note: Readings must be completed *before* the class that occurs in the assigned week (i.e., readings assigned for week 2 will be discussed *in* the week 2 class, thus they should be read *before*.).

Week 2:

Reading

- Wickham, H. (2015). “Teaching Safe-Stats, Not Statistical Abstinence”. In: *The American Statistician, Online Discussion*. https://nhorton.people.amherst.edu/mererenovation/17_Wickham.PDF.
- Wickham, H. and G. Grolemund (2016). “Preface”. In: *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. Ed. by H. Wickham and G. Grolemund. O’Reilly Media, pp. ix–2. <http://r4ds.had.co.nz/introduction.html>.
- Wickham, H. and G. Grolemund (2016). “Data Visualization with ggplot2”. In: *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. Ed. by H. Wickham and G. Grolemund. O’Reilly Media, pp. 3–34. <http://r4ds.had.co.nz/data-visualisation.html>.
- Wickham, H. and G. Grolemund (2016). “Workflow: Basics”. In: *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. Ed. by H. Wickham and G. Grolemund. O’Reilly Media, pp. 37–39. <http://r4ds.had.co.nz/workflow-basics.html>.
- Johnson, K. (2011). “Fundamentals of Quantitative Analysis”. In: *Quantitative Methods In Linguistics*. Ed. by K. Johnson. Wiley, pp. 1–33.

Additional resources

- Field Guide to the R Ecosystem (highly recommended)
- Data science and R: How do I start?
- GitHub for the userR
- Using Git and GitHub
- Getting help

Week 3:

Reading

- Johnson, K. (2011). “Patterns and Tests”. In: *Quantitative Methods In Linguistics*. Ed. by K. Johnson. Wiley, pp. 34–69.
- Wickham, H. and G. Grolemund (2016). “Data Transformation with dplyr”. In: *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. Ed. by H. Wickham and G. Grolemund. O’Reilly Media, pp. 43–73.
- R for excel users

Additional resources

- RMarkdown cheatsheet
- RMarkdown reference

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- Getting started with RMarkdown
- ggplot2 cheatsheet
- Secrets to a happy graphing life
- A Compendium of Clean Graphs in R



Week 4:

Reading

- Johnson, K. (2011). "Patterns and Tests". In: *Quantitative Methods In Linguistics*. Ed. by K. Johnson. Wiley, pp. 34–69.
- Wickham, H. and G. Grolemund (2016). "Workflow: Scripts". In: *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. Ed. by H. Wickham and G. Grolemund. O'Reilly Media, pp. 77–79.
- Wickham, H. and G. Grolemund (2016). "Exploratory Data Analysis". In: *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. Ed. by H. Wickham and G. Grolemund. O'Reilly Media, pp. 81–108.

Additional resources

- The tidyverse style guide
- reprex 1
- reprex 2
- Data structure basics
- Correlation game
- Standardizing and correlations

Week 5:

Reading

- Lewis-Beck, M. (1980). "Bivariate Regression: Fitting a Straight Line". In: *Applied Regression: An Introduction*. Ed. by M. Lewis-Beck. Sage University Paper Series on Quantitative Applications in the Social Sciences - 22. Newbury Park, CA: Sage, pp. 9–25. ISBN: 9781483381497.
- Wickham, H. and G. Grolemund (2016). "Workflow: Projects". In: *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. Ed. by H. Wickham and G. Grolemund. O'Reilly Media, pp. 111–116.
- Wickham, H. and G. Grolemund (2016). "Tidy Data with tidyr". In: *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. Ed. by H. Wickham and G. Grolemund. O'Reilly Media, pp. 147–168.

Additional resources

- tidyr tutorial
- Tidy data
- Non-tidy data
- Creating new variables with mutate and ifelse (advanced)

Week 6:

- Schroeder, L., D. Sjoquist, and P. Stephan (1986). “Linear Regression”. In: *Understanding Regression Analysis: An Introductory Guide*. Ed. by L. Schroeder, D. Sjoquist and P. Stephan. Sage University Paper Series on Quantitative Applications in the Social Sciences - 57. Newbury Park, CA: Sage, pp. 11–28. ISBN: 9780803927582.
- Lewis-Beck, M. (1980). “Bivariate Regression: Assumptions and Inferences”. In: *Applied Regression: An Introduction*. Ed. by M. Lewis-Beck. Sage University Paper Series on Quantitative Applications in the Social Sciences - 22. Newbury Park, CA: Sage, pp. 26–46. ISBN: 9781483381497.
- Wickham, H. and G. Grolemund (2016). “Data Import with readr”. In: *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. Ed. by H. Wickham and G. Grolemund. O’Reilly Media, pp. 125–145.
- Wickham, H. and G. Grolemund (2016). “R Markdown”. In: *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. Ed. by H. Wickham and G. Grolemund. O’Reilly Media, pp. 423–438.
- Slidify tutorial

Additional resources

- Fit-a-line
- Simple linear regression game
- Bivariate linear regression (shiny)

Week 7:

Reading

- Lewis-Beck, M. (1980). “Multiple Regression”. In: *Applied Regression: An Introduction*. Ed. by M. Lewis-Beck. Sage University Paper Series on Quantitative Applications in the Social Sciences - 22. Newbury Park, CA: Sage, pp. 47–74. ISBN: 9781483381497.
- Schroeder, L., D. Sjoquist, and P. Stephan (1986). “Multiple Linear Regression”. In: *Understanding Regression Analysis: An Introductory Guide*. Ed. by L. Schroeder, D. Sjoquist and P. Stephan. Sage University Paper Series on Quantitative Applications in the Social Sciences - 57. Newbury Park, CA: Sage, pp. 29–35. ISBN: 9780803927582.
- xaringan tutorial 1
- xaringan tutorial 2

Additional resources

- R Markdown presentations
- Model choices/interactions
- Diagnostics for simple linear regression

Week 8:

Reading

- Schroeder, L., D. Sjoquist, and P. Stephan (1986). “Problems and Issues of Linear Regression”. In: *Understanding Regression Analysis: An Introductory Guide*. Ed. by L. Schroeder, D. Sjoquist and P. Stephan. Sage University Paper Series on Quantitative Applications in the Social Sciences - 57. Newbury Park, CA: Sage, pp. 65–80. ISBN: 9780803927582.

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- Berry, W. and S. Feldman (1985). "Specification Error". In: *Multiple Regression in Practice*. Ed. by W. Berry and S. Feldman. Sage University Paper Series on Quantitative Applications in the Social Sciences - 50. Newbury Park, CA: Sage, pp. 18–25. ISBN: 9780803920545.
- Berry, W. and S. Feldman (1985). "Multicollinearity". In: *Multiple Regression in Practice*. Ed. by W. Berry and S. Feldman. Sage University Paper Series on Quantitative Applications in the Social Sciences - 50. Newbury Park, CA: Sage, pp. 37–50. ISBN: 9780803920545.
- papaja documentation: Read documentation, install package and play with demo example.

Additional resources

- Multicollinearity
- papaja: Reproducible APA manuscripts with R Markdown (skim)
- More papaja
- R Markdown with papaja (video pt. 1)
- R Markdown with papaja (video pt. 2)

Spring break (highly recommended walk through):

Reading

- Wickham, H. and G. Grolemund (2016). "Model Basics with modelr". In: *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. Ed. by H. Wickham and G. Grolemund. O'Reilly Media, pp. 345–372.
- Wickham, H. and G. Grolemund (2016). "Model Building". In: *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. Ed. by H. Wickham and G. Grolemund. O'Reilly Media, pp. 375–396.

Week 9:

Reading

- Hardy, M. (1993). "Creating Dummy Variables". In: *Regression with Dummy Variables*. Ed. by M. Hardy. Sage University Paper Series on Quantitative Applications in the Social Sciences - 93. Newbury Park, CA: Sage, pp. 7–17. ISBN: 9780803951280.
- Hardy, M. (1993). "Using Dummy Variables as Regressors". In: *Regression with Dummy Variables*. Ed. by M. Hardy. Sage University Paper Series on Quantitative Applications in the Social Sciences - 93. Newbury Park, CA: Sage, pp. 18–28. ISBN: 9780803951280.
- Hardy, M. (1993). "Assessing Group Differences in Effects". In: *Regression with Dummy Variables*. Ed. by M. Hardy. Sage University Paper Series on Quantitative Applications in the Social Sciences - 93. Newbury Park, CA: Sage, pp. 29–63. ISBN: 9780803951280.

Additional resources

- Model choices/interactions
- The power of multiple regression
- StackOverflow question w/ good answer
- Dummy variables (video)

Week 10:

Reading

- Hardy, M. (1993). “Alternative Coding Schemes for Dummy Variables”. In: *Regression with Dummy Variables*. Ed. by M. Hardy. Sage University Paper Series on Quantitative Applications in the Social Sciences - 93. Newbury Park, CA: Sage, pp. 64–74. ISBN: 9780803951280.

Additional resources

- Coding systems
- Coding schemes for categorical variables in regression

Week 11 (exam week):

Week 12:

Reading

- McElreath, R. (2015). “Big Entropy and the Generalized Linear Model”. In: *Statistical Rethinking: A Bayesian Course with Examples in R and Stan*. Ed. by R. McElreath. Chapman & Hall/CRC Texts in Statistical Science. CRC Press, pp. 267–290. ISBN: 9781482253481.
- Gelman, A. and J. Hill (2007). “Generalized linear models”. In: *Data Analysis Using Regression and Multi-level/Hierarchical Models*. Ed. by A. Gelman and J. Hill. Analytical Methods for Social Research. Cambridge University Press, pp. 109–134. ISBN: 9780521686891.

Additional resources

- Generalised Linear Models in R
- Shiny GLM
- When to use logistic regression
- Why regular regression does NOT work
- What it (multivariate) logistic regression

Week 13:

Reading

- Gelman, A. and J. Hill (2007). “Logistic regression”. In: *Data Analysis Using Regression and Multi-level/Hierarchical Models*. Ed. by A. Gelman and J. Hill. Analytical Methods for Social Research. Cambridge University Press, pp. 79–108. ISBN: 9780521686891.

Additional resources

- What a Multivariate Logistic Regression Data Set Looks Like: An Example
- A little about logs
- About odds and odds ratios
- Understanding Logistic Regression Output
- Sample sizes